Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Course : B.Tech
Code : 103403

Subject: Mathematics -III (Probability and Statistics)

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin
(ii) There are **NINE** questions in this paper
(iii) Attempt **FIVE** questions in all
(iv) Question No. 1 is compulsory

Q.1 Choose the correct answer of the following (any seven question only): [2 x 7 = 14]

(a) In a bulb factory, machines A, B and C manufactures 60%, 30% and 10% bulbs respectively. 1%, 2% and 3% of the bulbs produced respectively by A, B and C are found to be defective. A bulb is picked up at random from the total production and found to be defective. Find the probability that this bulb was produced by the machine A.

(i) $\frac{1}{3}$ (ii) $\frac{2}{5}$ (iii) $\frac{1}{5}$ (iv) $\frac{3}{4}$

(b) A random variable X has the following probability distribution:

x	-2	-1	0	1	2	3
P(x)	0.1	k	0.2	2k	0.3	k

The value of k is:

(i) .001 (ii). 05 (iii) 0.1 (iv) 0.02

(c) Which of the following is not a measure of Dispersion?

(i) Mean Deviation
(ii) Standard Deviation
(iii) Quartile Deviation
(iv) Average Deviation from mean.

(d) The value of variance of a discrete random variable X is given by:

(i) $V(X) = E(X^2) - \{E(X)\}^2$ (ii) $V(X) = \int x^2 f(x)dx$
(iii) $V(X) = P(X = x)$ (iv) $V(X) = \sum x^3 P(X = x)$

(e) For a Negative skewed frequency distribution curve, the third central moment:

(i) $\mu_3 > 0$ (ii) $\mu_3 < 0$
(iii) $\mu_3 = 0$ (iv) μ_3 *does not exist*.

(f) For a Symmetrical distribution, the coefficient of Skewness:

(i) $\alpha_3 = 1$ (ii)) $\alpha_3 = 3$
(iii) $\alpha_3 = 0$ (iv)) $\alpha_3 = -1$

(g) If X and Y are two variates, these can be at most:

(i) One regression line (ii) Two regression lines
(iii) Three regression lines (iv) an infinite number of regression lines

(h) If X follows a binomial distribution with parameters n=20and p=0.4, what is the expected value E(X)?

(i) 8 (ii) 10
(iii) 12 (iv) 6

(i) Probability mass function for a binomial distribution with usual notations is:

(i) $\binom{n}{k}p^{k-1}q$ (ii) $\binom{n}{k}p^k q$
(iii) $\binom{n}{k}p^k q$ (iv) $\binom{n}{k}p^k q^{n-k}$

(j) The T-test is used to compare:

(i) The means of two groups (ii) The medians of two groups
(iii) The variances of two groups (iv) The standard deviations of two groups.

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Q.2 (a) Two cards are drawn without replacement from a pack of 52 cards. Find the probability that: (i) both are red (ii) the first is a king and the second is an ace. [7]

(b) In a bulb factory, machines A, B and C manufactures 40%, 40% and 20% bulbs respectively. 4%, 2% and 3% of the bulbs produced respectively by A, B and C are found to be defective. A bulb is picked up at random from the total production and found to be defective. Find the probability that this bulb was produced by the machine A [7]

Q.3 (a) Find the Expectation of number of heads in three tosses of fair coin. Also find (i) Mean (ii) $E(X^2)$ (iii) Variance. [7]

(b) The diameter X of an electric cable is assumed to be continuous random variable with probability density function is: [7]

$f(x) = 6x(1-x),\ 0 \le x \le 1.$

(i)Check that above is probability density function.

(ii) Compute $P(0.5 \le x \le 0.75)$.

Q.4 (a) In a normal distribution 31 % of the items are under 45 and 8 % are over 64. Find the mean and variance of the distribution. [7]

$[Given f(1.4) = 0.42, f(0.5) = 0.19]$.

(b) The length of telephone conversion is an exponential variate with mean 3 minutes. Find the probability that call (a) end less than 3 min. (b) takes between 3 to 5 min. [7]

Q.5 (a) Find the value of k if $f(x, y) = k(1-x)(1-y)$ for $0 < x, y < 1$, is to be joint density function. [7]

(b) If the joint p.d.f. of (X, Y) is $f(x, y) = 6e^{-2x-3y}$, $x \ge 0, y \ge 0$, find the conditional density of Y given X. [7]

Q.6 (a) Fit a second degree parabola to the following data: [4]

x	0	1	2	3	4
y	1	4	10	17	30

(b) A manufacturer claims that only 4% of his products supplied by him are defective. A random sample of 600 products contains 36 defectives. Test the claim of the manufacturer at 5% level. [10]

Q.7 Samples of sizes 10 and 14 were taken from two normal population of S.D. 3.5 and 5.2.The sample means were Found to be 20.3 and 18.6. Test whether the means of the population are same at 5% level of significance. (Given, t at 5% level of significance is 2.07). [14]

Q.8 The number of yeast cells counted in a haemocytometer is compared to the theoretical value is given below. Does the experimental result support the theory? χ^2(6-1=5 d. f. at 5 %) = 11.070. [14]

No. of Yeast cells in the square	Observed Frequency	Expected Frequency
0	103	106
1	143	141
2	98	93
3	42	41
4	8	14
5	6	5

Q.9 Write short notes on ***any two*** of the following:- [7 x 2 = 14]

(a) marginal probability distribution

(b) Z- Test for significance of difference of two means in case of Large samples.

(c) Chi-square test for goodness of fit.

(d) Large sample test for Difference of standard deviation.

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